

UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The study of Higgs properties in the four-lepton final state at CMS

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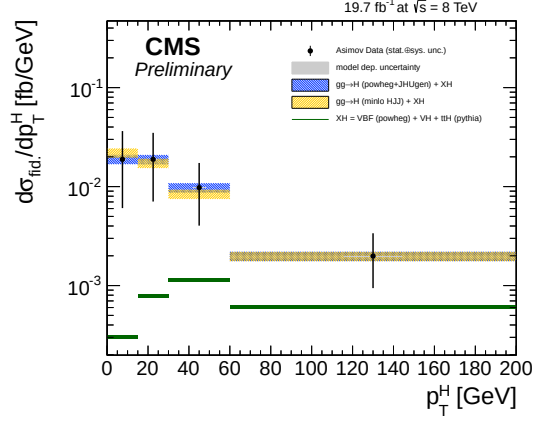
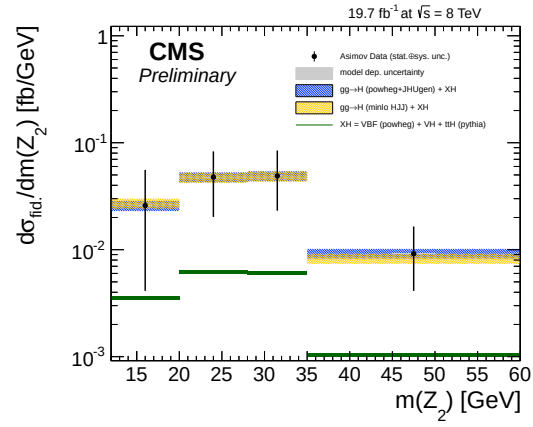
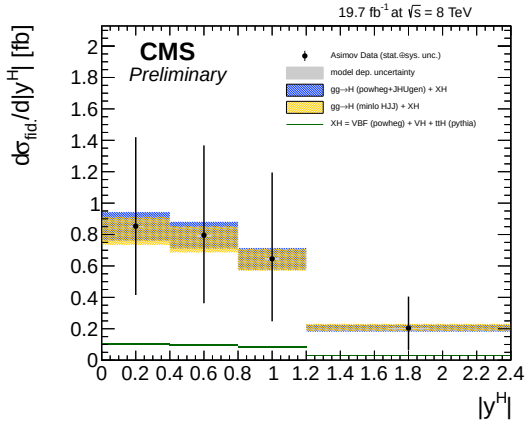
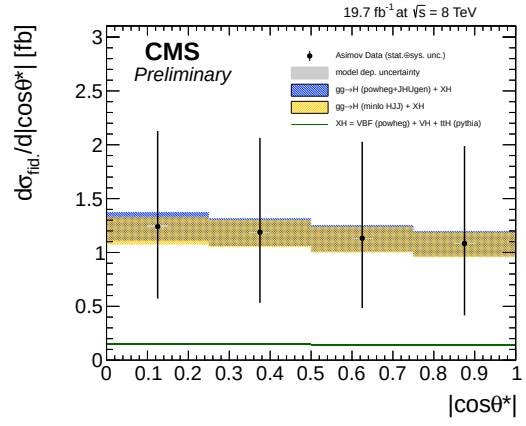
- 1 Results of the differential cross section measurement for different observables when the asimov dataset is generated assuming pure ZH production with cross section equal to the total SM cross section. The prediction for this pure ZH model is shown in red. The central value of the measurement is obtained using SM efficiencies with the $gg \rightarrow H$ prediction from POWHEG+JHUGEN with $m(H) = 125$ GeV, and the error bar represents the combined statistical and systematic uncertainty. The grey box represents the additional uncertainty incurred when using different models to unfold the observed data distribution back to particle level. 4

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*Dedicated to my family who endured missed birthdays,
missed festivals and countless days alone so that I may
make history*

Appendix

.1 Results with model dependance uncertainty constrained with experimental results

(a) $p_T(H)$ (b) $m(Z_2)$ (c) $|y(H)|$ (d) $|\cos\theta^*|$ 